

Deploy a Web Application to Kubernetes using Minikube.

STEP 1 - Start Minikube

minikube start

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg7$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
🚧 Pulling base image v0.0.48 ...
🔧 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image docker.io/kubernetes/dashboard:v2.7.0
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetes/metrics-scraper:v1.0.8
```

STEP 2 - Check Minikube Status

minikube status

You should see:

host: Running

kubelet: Running

apiserver: Running

kubeconfig: Configured

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

STEP 3 - If Minikube is not running, restart it

minikube delete

minikube start --driver=docker

STEP 4 - Create the Deployment file (nginx-deployment.yaml)

Create a new file:

nginx-deployment.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

name: nginx-deployment

spec:

replicas: 2

selector:

matchLabels:

app: nginx

template:
metadata:
labels:
app: nginxspec:
containers:
- name: nginx
image: nginx:latest
ports:
- containerPort: 80

STEP 5 - Apply the deployment

kubectl apply -f nginx-deployment.yaml

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

STEP 6 - Check Deployment & Pods

kubectl get deployments

kubectl get pods

You should see 2 pods running.

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    0/2     2             0           10s
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg7$ kubectl get pods
NAME                                READY   STATUS             RESTARTS   AGE
nginx-deployment-6f9664446b-2mpqr   1/1     Running            0           16s
nginx-deployment-6f9664446b-wljgq   0/1     ContainerCreating  0           16s
```

STEP 7 -Expose the Deployment as a Service

kubectl expose deployment nginx-deployment \

--type=NodePort \

--port=80

This creates a service that exposes Nginx to your system.

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg7$ kubectl expose deployment nginx-deployment --
type=NodePort --port=80
service/nginx-deployment exposed
```

STEP 8 - Check Services

kubectl get services

A service named nginx-deployment should appear .

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg7$ kubectl get services
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1    <none>        443/TCP          8m12s
nginx-deployment    NodePort    10.101.24.25 <none>        80:30269/TCP     11s
```

STEP 9 - Access the Webpage

Minikube provides a shortcut to open the app:

minikube service nginx-deployment

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30269

🌈 Opening service default/nginx-deployment in default browser...

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/pg7$ Opening in existing browser session.
```

This will open the browser showing the Nginx Welcome Page. STEP 10 — Clean Up (Optional)

Delete all pods

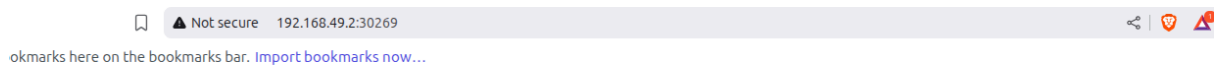
```
kubectl delete pods --all
```

Delete all services

```
kubectl delete svc --all
```

Delete all deployments

```
kubectl delete deployments --all
```



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.